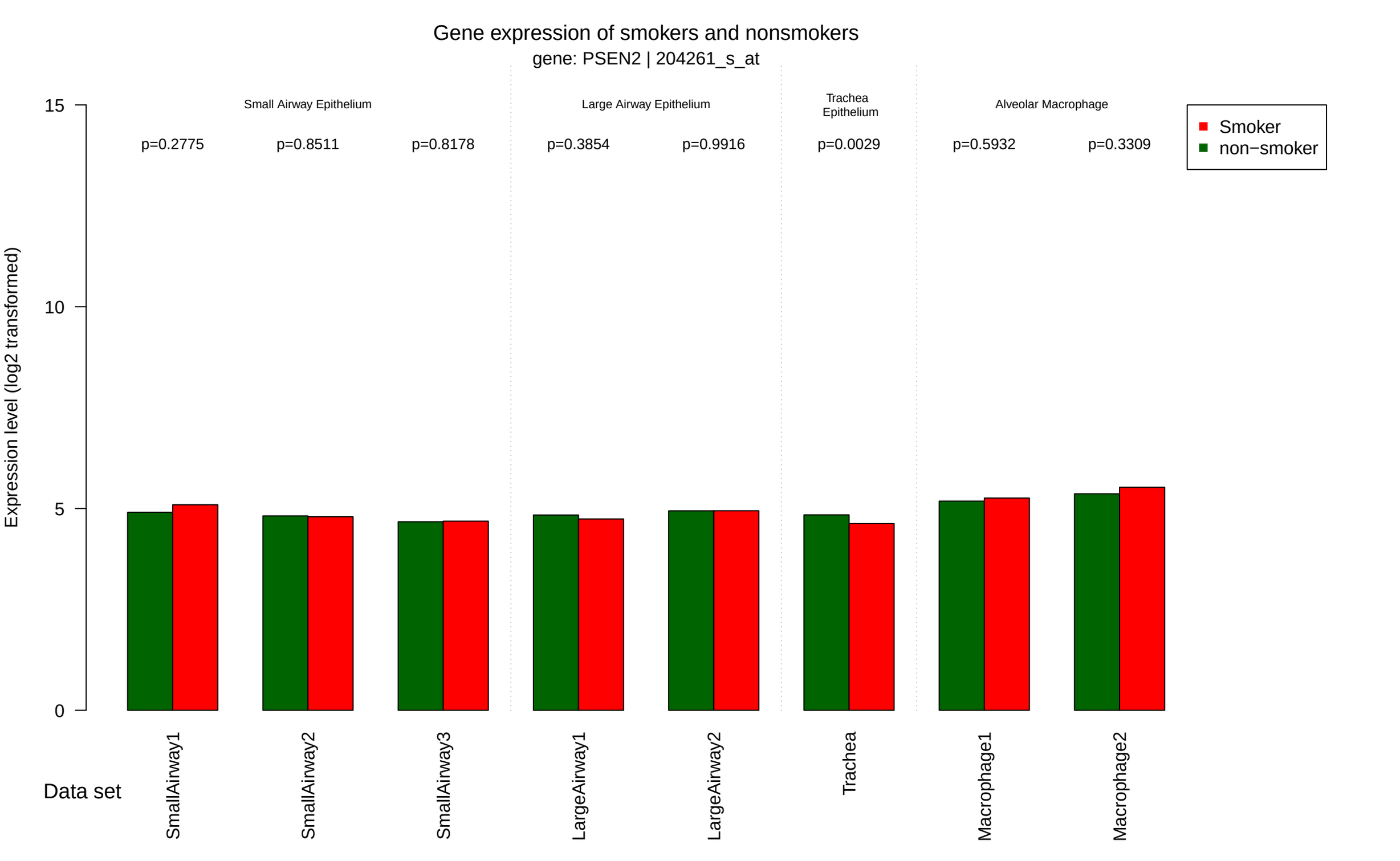
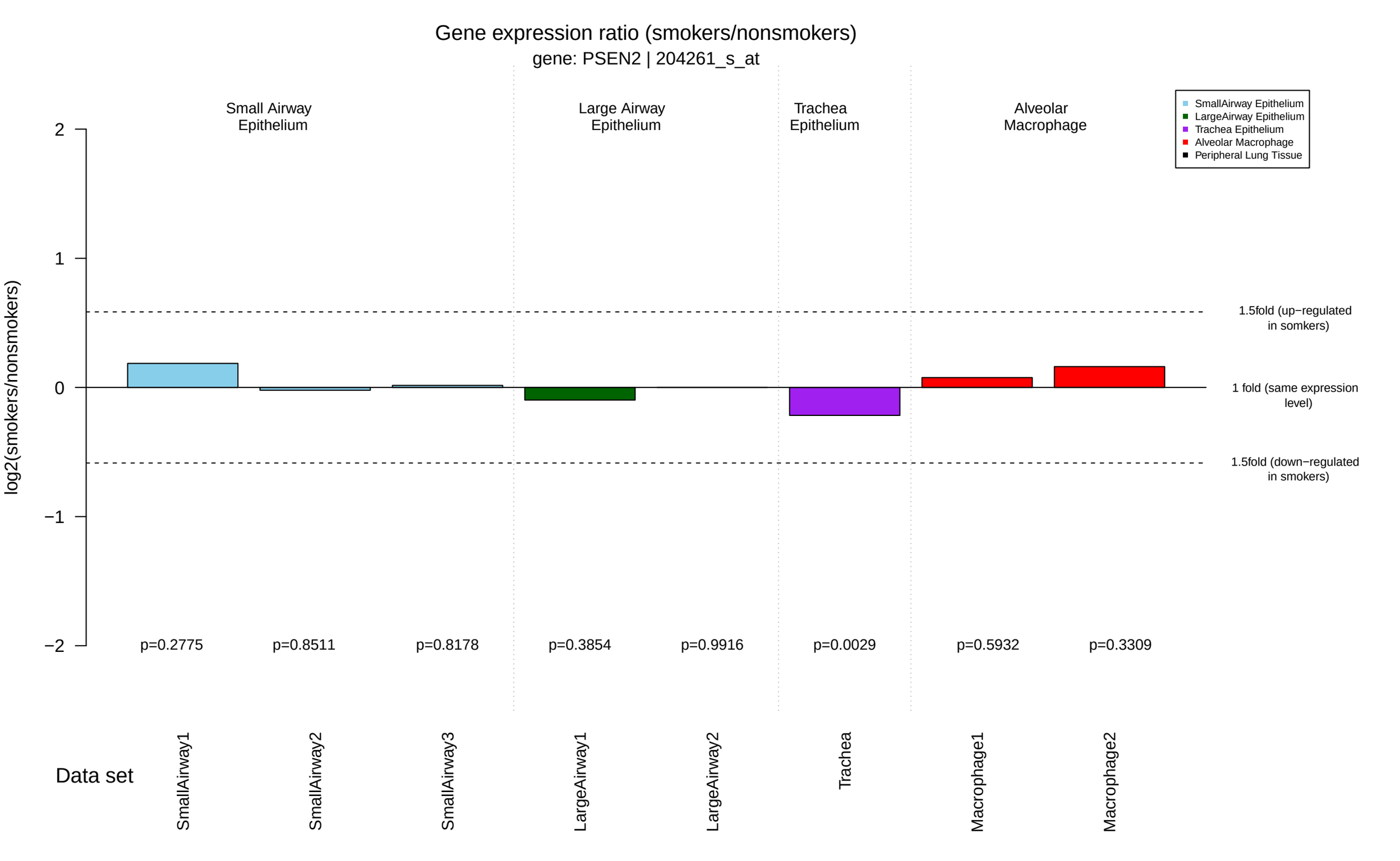
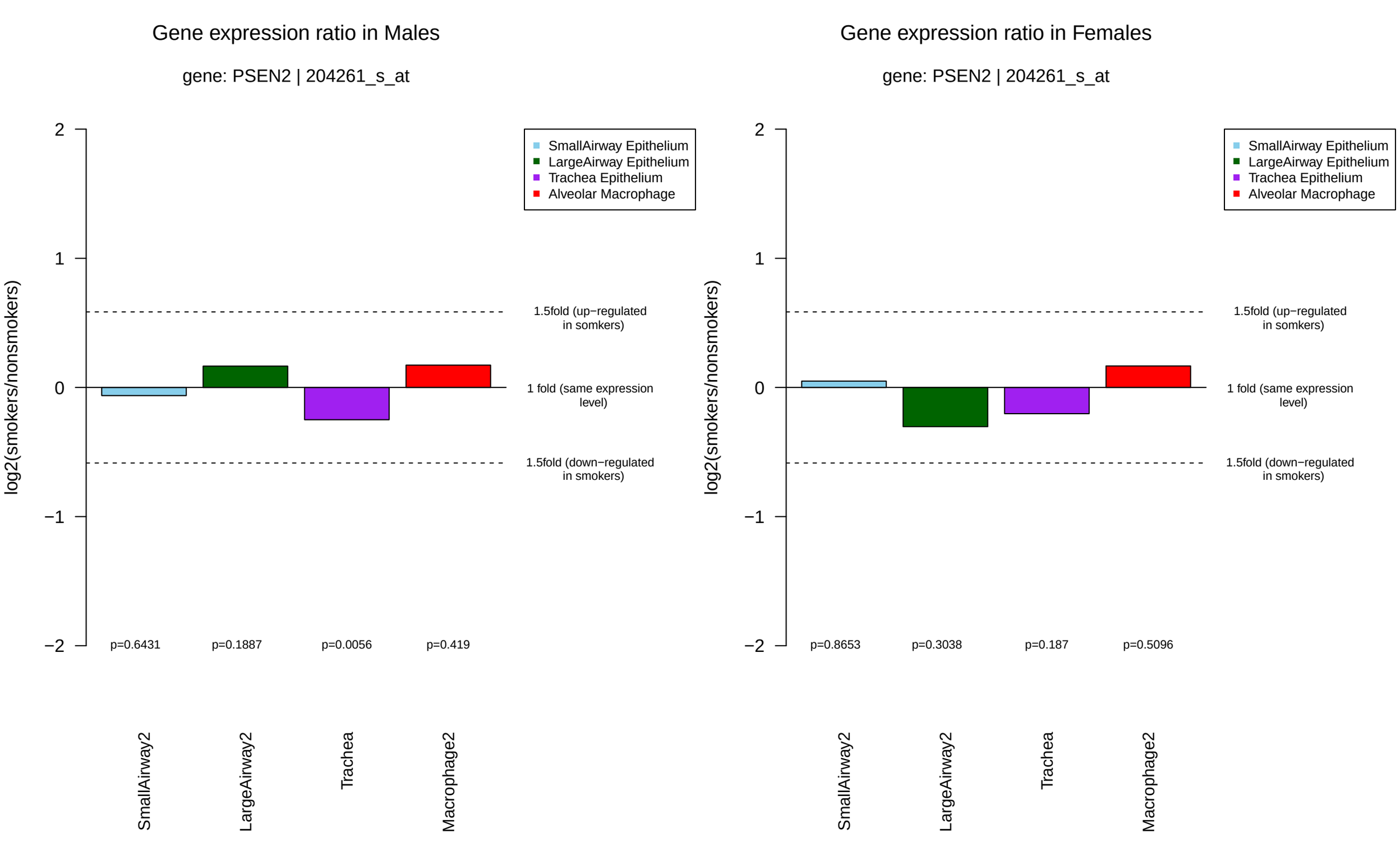
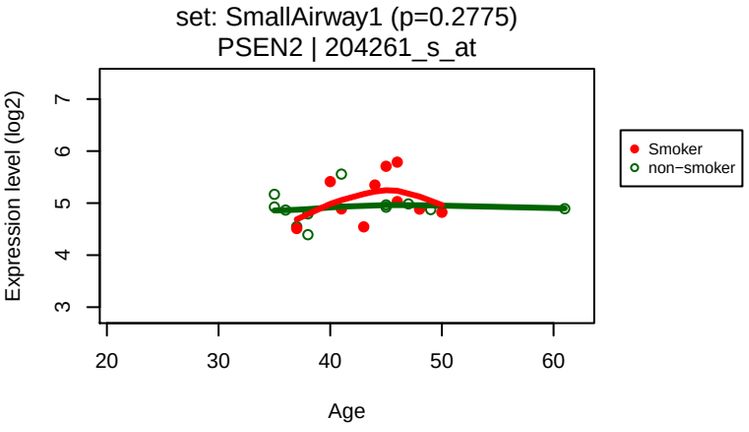
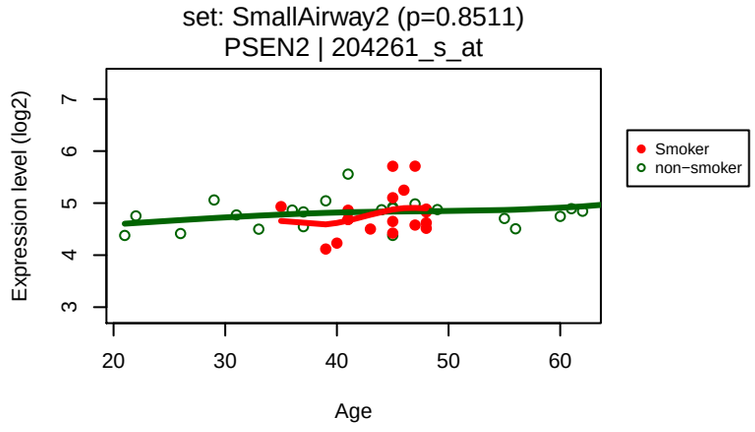
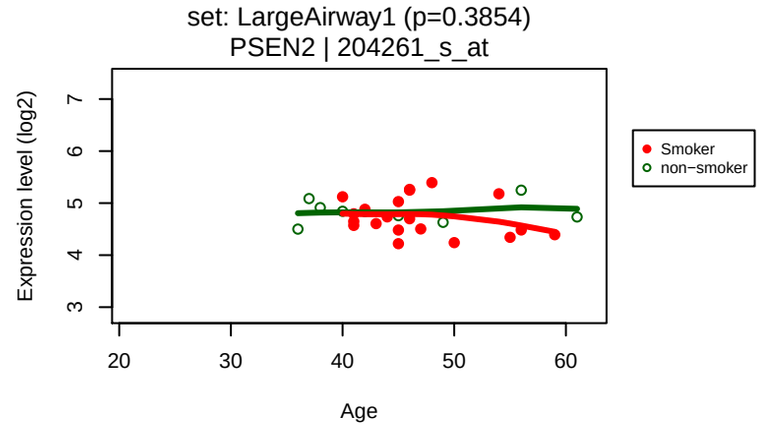

Gene expression of smokers and nonsmokers
gene: PSEN2 | 204261_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2775
p=0.8511
p=0.8178
p=0.3854
p=0.9916
p=0.0029
p=0.5932
p=0.3309
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

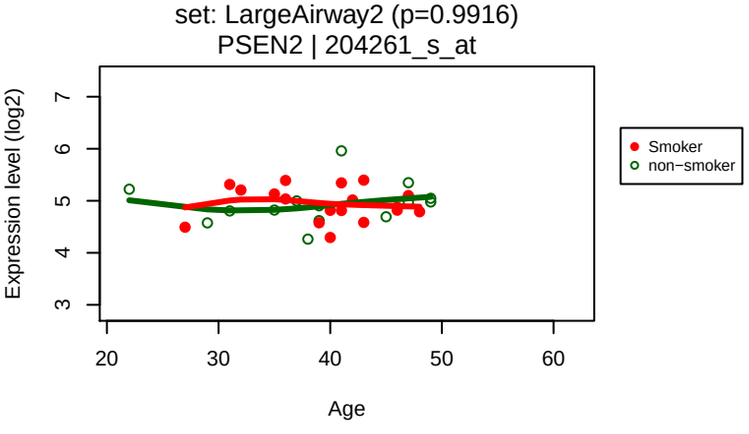

Gene expression ratio (smokers/nonsmokers)
gene: PSEN2 | 204261_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2775
p=0.8511
p=0.8178
p=0.3854
p=0.9916
p=0.0029
p=0.5932
p=0.3309
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

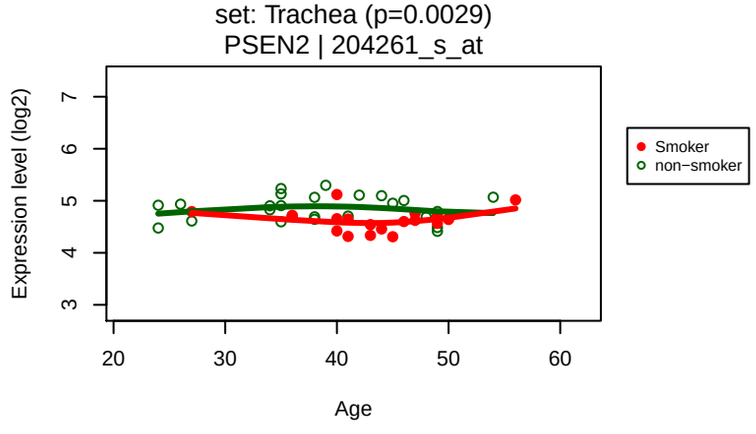

Gene expression ratio in Males
gene: PSEN2 | 204261_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6431
p=0.1887
p=0.0056
p=0.419
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSEN2 | 204261_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8653
p=0.3038
p=0.187
p=0.5096
SmallAirway2
LargeAirway2
Trachea
Macrophage2

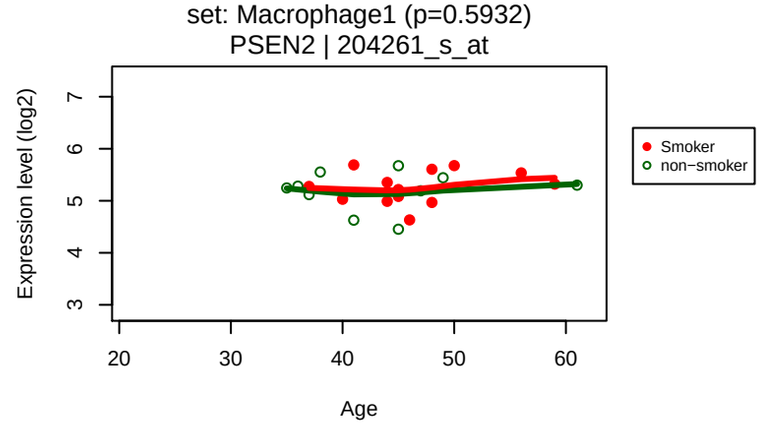

set: SmallAirway1 (p=0.2775)
PSEN2 | 204261_s_at
Expression level (log2)
Age
Smoker
non−smoker

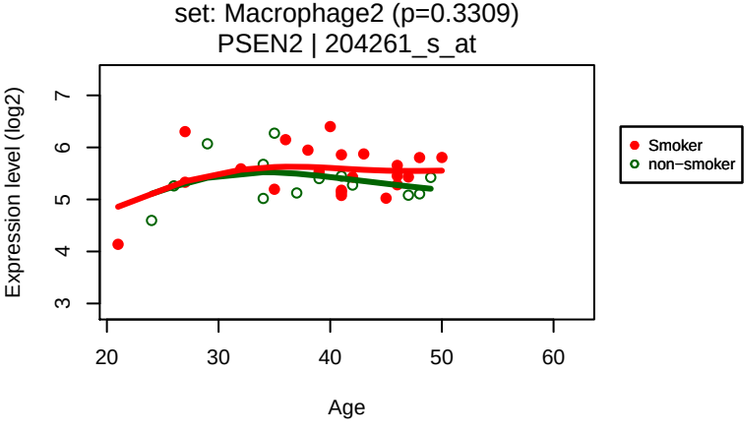

set: SmallAirway2 (p=0.8511)
PSEN2 | 204261_s_at
Expression level (log2)
Age
Smoker
non−smoker

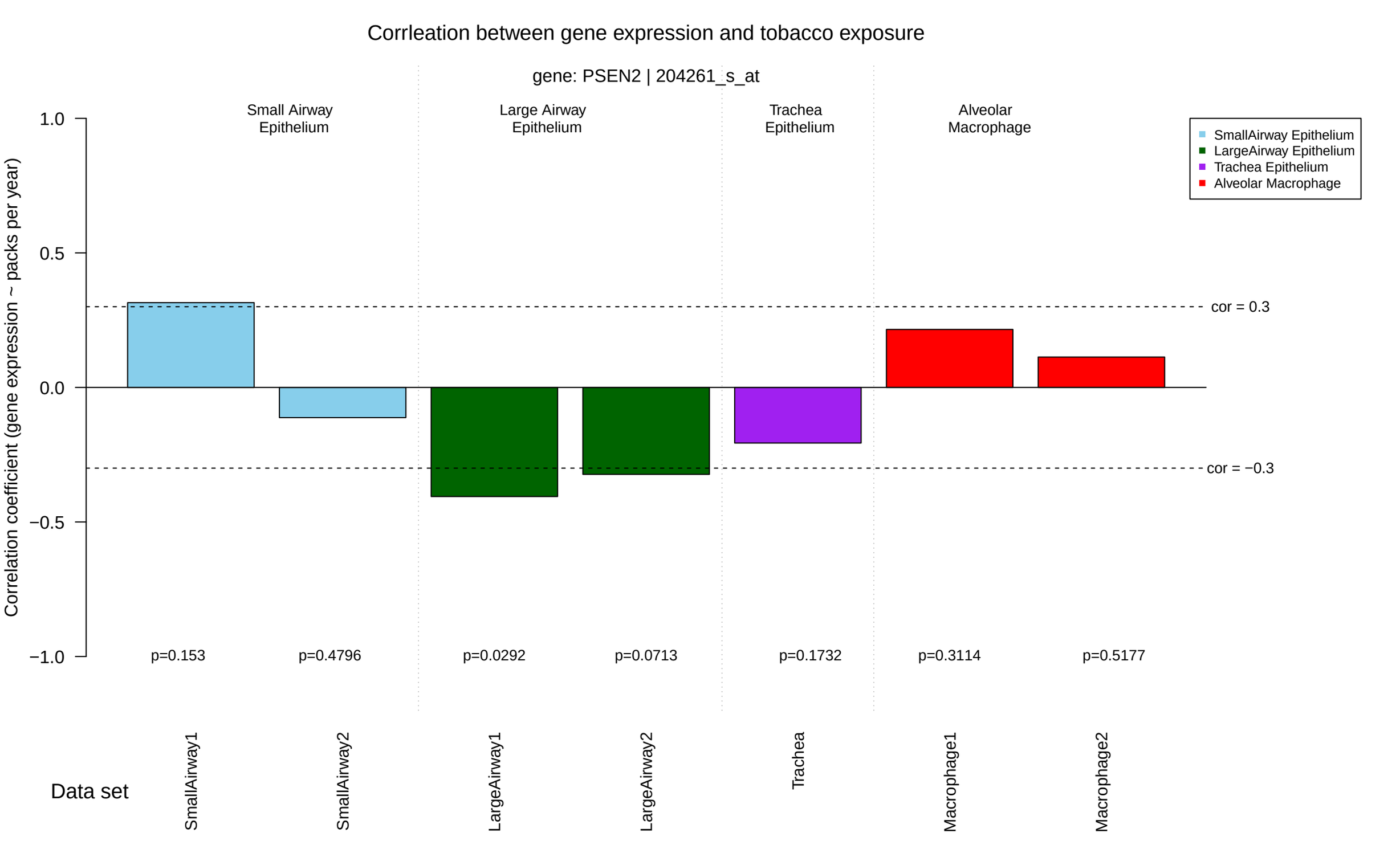

set: LargeAirway1 (p=0.3854)
PSEN2 | 204261_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9916)
PSEN2 | 204261_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0029)
PSEN2 | 204261_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5932)
PSEN2 | 204261_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3309)
PSEN2 | 204261_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSEN2 | 204261_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.153
p=0.4796
p=0.0292
p=0.0713
p=0.1732
p=0.3114
p=0.5177
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2